

Network Cable Configuration & Linux Physical Network Setup

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ENSE 353 – Lab 2

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Objective

(What did we do?):

Construction of a Cat5e network cable & a bare-metal Linux system host with a reachable Apache web server.

Steps

(How did we do it?)

Part 1: Crimping cable

Standard used:

T-568A

Cable color distribution:

Left WG, G, WO, B, WB, O, WBr, Br Right

A horizontal blue line with a right-pointing arrow at its end, representing the sequence of cable colors from left to right.

Connector Type:

RJ45 Slightly wider than a telephone line
Cat5e written on the cable

Part 2: Using Cat5e to Setup Linux Network

1. Computer plugged all connected. When on type:

`shutdown -r now`

2. Check network

`ip a`

3. Go to network manager

`Nmtui`

4. Activate a network connection

`enp0s31f6`

Click 'Activate' and repeat 'ip a' to check network.
You should not see your inet address on (2).

5. Test connection with a ping

`ping google.ca`

You should see pinging by the loading lines (testing connection)

6. Keep server up to date

`dnf upgrade --refresh`

7. Show the current kernel

`uname -r`

8. Reboot, so new kernel is loaded and set as default

9. If the network does not automatically connect, 'nmtui' & edit connection & set it as automatic.

10. Download emacs text editor

`dnf install emacs-nox`

11. Download apache

`dnf install httpd`

12. Check the status of what is going on

`systemctl status httpd`

13. Start your server

`system start httpd`

14. Check status again (Step 12)

15. Set the PC to start server every time it boots in case of a power outage. This creates a symlink.

```
systemctl enable httpd
```

16. Never communicate over TelNet, it is not secure. We are using it as a tool.

```
dnf install telnet
```

Apache server files on Port80

http on port80 is how the browser connects to servers

```
'telnet 127.0.0.1 port 80'
```

It will boot you out if left stagnant.

17. Type the IP address in a browser to go to server.

```
<IP_Address>:port8080
```

18. Should have no connection, check firewall:

```
firewall-cmd --state
```

& check

```
firewall-cmd --list-all
```

19. Allow http on firewall

```
Firewall-cmd --permanent --zone=public --add-service=http
```

20. Refresh firewall

```
firewall-cmd --reload
```

21. Check again the list of all

```
firewall-cmd --list-all
```

22. Make a file on editor

```
emacs index.html
```

23. Save & quit

```
(ctrl + x) + (ctrl + c) + y
```

Outcome

(Why did we do it?)

End-to-end connection established.

Demonstration in terminating physical network media, managing network interface and how to security route HTTP traffic through a system firewall.